

Day 1 — Introduction to Python

Detailed Seminar Notes | Definition • Features • Uses • Installation • First Program • Rules • Viva

Day 1 is the starting point of your Python journey. In this day, the goal is to understand **what Python is**, **why it is popular**, **where it is used**, how to **install and run Python**, and how to write the **first Python program**.

1) What is Python?

Python is a **high-level, interpreted, general-purpose programming language**. It was created by **Guido van Rossum** and first released in **1991**. Python is designed to be easy to read, easy to learn, and powerful enough to build real applications.

It is called a **general-purpose language** because it can be used in many areas such as **web development, automation, data science, machine learning, scripting, software development, and cybersecurity**.

2) Why is Python popular?

- **Simple syntax** — Python code looks clean and close to English.
- **Beginner friendly** — easy to start even if you are new to programming.
- **Less code** — many tasks can be done in fewer lines compared to some other languages.
- **Large community** — lots of tutorials, libraries, and support are available.
- **Powerful libraries** — libraries like **NumPy, Pandas, Matplotlib, TensorFlow**, and **Django** make development easier.
- **Used in real industry** — Python is widely used by companies and developers.

3) Important definitions related to Python

Term	Meaning
Programming Language	A language used to give instructions to a computer.
High-Level Language	A language that is easy for humans to read and write.
Interpreted Language	A language where code is executed line by line by an interpreter.
Compiler	A program that converts the entire source code into machine code at once.
Interpreter	A program that translates and executes code line by line.
Syntax	The set of rules used to write valid code in a programming language.

4) Features of Python

- **Easy to read and write** — Python uses simple words and clean structure.
- **Interpreted** — code is executed line by line.
- **Platform independent** — a Python program can run on Windows, Linux, and macOS with little or no change.
- **Object-oriented** — Python supports object-oriented programming.
- **Open source** — Python is free to use and modify.
- **Large standard library** — many built-in modules are available.

- **Extensible and embeddable** — Python can work with other languages and systems.

5) Applications of Python

Python is used in many domains. A few important applications are listed below.

Area	How Python is used
Web Development	Building websites and backend applications using frameworks like Django and Flask.
Data Science	Data analysis, cleaning, and visualization using Pandas, NumPy, and Matplotlib.
Machine Learning / AI	Model building and AI development using scikit-learn, TensorFlow, and PyTorch.
Automation / Scripting	Automating repetitive tasks like file handling, emails, reports, and scraping.
Game Development	Building simple games and game logic.
Cybersecurity	Writing scripts for testing, automation, and analysis.
Desktop Applications	Creating software applications with Python libraries.

6) Python versions: Python 2 vs Python 3

Today, we use **Python 3**. Python 2 is old and no longer officially supported. Whenever you learn or write programs now, use **Python 3**.

Python 2	Python 3
Older version, officially discontinued.	Current version used today.
Some syntax is different.	Cleaner and improved syntax.
Not recommended for new learning.	Recommended for all new learners and projects.

7) Installing Python

To write and run Python programs, Python must be installed on your computer.

Basic steps:

- Go to the official Python website and download Python 3.
- Run the installer.
- During installation, make sure to tick **Add Python to PATH**.
- Complete the installation.
- Open terminal / command prompt and type **python --version** or **py --version** to verify installation.

8) Ways to write Python code

Tool / Method	Use
Python IDLE	Simple editor that comes with Python installation.
VS Code	Popular code editor with Python support.
Jupyter Notebook	Useful for learning, notes, and data science.
Online Python Compiler	Good for quick testing without installation.

9) First Python program

The most basic Python program is printing text on the screen.

```
print("Hello, World!")
```

Output: Hello, World!

The **print()** function is used to display output on the screen. In this program, the string **Hello, World!** is printed.

10) Understanding the first program

Part	Meaning
print	A built-in Python function used to display output.
()	Parentheses used to pass data to the function.
"Hello, World!"	A string value enclosed in quotes.

11) More examples of print()

```
print("Python")
print("Welcome to programming")
print(10)
print(5 + 3)
```

This shows that **print()** can display text, numbers, and even the result of expressions.

12) Rules for writing Python programs

- Python is **case sensitive**. For example, **print** and **Print** are not the same.
- Use correct **syntax** — parentheses, quotes, and colons must be placed properly.
- Indentation is very important in Python, especially in conditions, loops, and functions.
- Strings must be enclosed in **single quotes** or **double quotes**.
- Comments can be written using the **#** symbol.
- Use meaningful names and clean formatting.

13) Comments in Python

Comments are notes written inside the code for explanation. Python ignores comments during execution.

```
# This is a comment
print("Hello")
```

Comments are useful to explain what the program is doing.

14) Difference between compiler and interpreter

Compiler	Interpreter
Translates the whole program at once.	Translates and executes one line at a time.
Usually gives errors after full compilation.	Can stop at the line where an error occurs.
Common in languages like C/C++.	Python mainly uses an interpreter approach.

15) Advantages of Python for beginners

- Readable syntax makes it easy to understand programs.
- You can focus on logic instead of complex syntax.
- Python is useful both for academics and industry.
- It is a strong base for future topics like **data science**, **web development**, **automation**, and **AI**.

16) Common beginner mistakes on Day 1

- Writing **Print()** instead of **print()**.
- Forgetting quotes around text values.
- Using mismatched quotes like opening with single quote and closing with double quote.
- Not installing Python correctly or forgetting to add it to PATH.
- Typing code in the terminal incorrectly.

17) Simple practice programs for Day 1

Program 1: Print your name

```
print("Adi")
```

Program 2: Print your college name

```
print("My college name is XYZ")
```

Program 3: Print a sentence and a number

```
print("Python is easy")  
print(100)
```

Program 4: Print the result of addition

```
print(10 + 20)
```

Program 5: Use a comment

```
# This program prints a welcome message  
print("Welcome to Python")
```

18) Day 1 seminar explanation format

If you are presenting Day 1 in a seminar, you can explain it in this order:

- Start with the definition of Python.
- Tell who created Python and why it is popular.
- Explain features of Python.
- Mention applications of Python in real life.
- Explain the difference between compiler and interpreter.
- Show how to install Python and where to write code.
- Run the first program **print("Hello, World!")**.
- Explain **print()**, strings, and comments.

19) Viva / interview questions for Day 1

- What is Python?
- Who developed Python?
- Why is Python called a high-level language?
- What do you mean by interpreted language?
- What is the difference between compiler and interpreter?
- Why is Python popular among beginners?
- What is the use of print()?
- What is a comment in Python?
- What are the applications of Python?
- Which Python version should we use today?

20) Short seminar summary

Python is a **simple, powerful, and beginner-friendly programming language**. It is interpreted, easy to read, and used in many areas like web development, data science, AI, and automation. Day 1 mainly focuses on understanding what Python is, its features, where it is used, how to install it, and how to write the first program using **print()**.

21) Quick revision box

Topic	Key point
Python	A high-level, interpreted, general-purpose programming language
Creator	Guido van Rossum
Popular because	Simple syntax, beginner friendly, powerful libraries
Used in	Web, AI, Data Science, Automation, Scripting
First program	<code>print("Hello, World!")</code>
Comment	Written using # and ignored by Python